

Chapter 28 - The M68K Processor

The M68K is the 16/32-bit processor that replaced the 6502 and the Z80 in the late 1980s machines: the Atari ST, the Commodore Amiga, the early Apple Macintosh, and the Sega Mega Drive. Intuition Engine implements the 68020 variant, which adds bit field operations, scaled indexing, full extension words, and a 32-bit multiply-divide on top of the original 68000 instruction set.

It has eight 32-bit data registers, eight 32-bit address registers, a flat 32-bit address space, and a supervisor/user mode split. Unlike the smaller chips, it sees Intuition Engine's MMIO map directly without any banking.

28.1 User-mode programmer's model

Group	Registers
Data registers	D0-D7, each 32 bits
Address registers	A0-A7, each 32 bits (A7 is the stack pointer)
Program counter	PC, 32 bits
Condition code reg	CCR, low byte of the status register

Each data register holds an 8-bit, 16-bit, or 32-bit value (byte/word/long). The size is selected by a suffix on the mnemonic: `MOVE.B`, `MOVE.W`, `MOVE.L`. Operations on the smaller sizes leave the unused high bits of the register unchanged.

Address registers always hold a 32-bit value. Sign-extension from 16-bit operations fills the whole register. The unique A7 (also called SP) is the active stack pointer and switches between user and supervisor copies on mode change.

The CCR has five condition code bits arranged X N Z V C (low to high) within an 8-bit byte:

Bit	Name	Meaning
4	X	Extend - copy of C for arithmetic; preserved for some shifts
3	N	Negative - bit 7/15/31 of the result
2	Z	Zero - result was zero
1	V	Overflow - signed overflow occurred
0	C	Carry - unsigned overflow / borrow

28.2 Supervisor-mode model

In supervisor mode, the user model is augmented with:

- The full 16-bit status register SR, of which the low byte is the CCR. The high byte holds:
- bit 15 T1, bit 14 T0: trace mode selectors
- bit 13 S: supervisor flag (1 = supervisor)
- bit 12 M: master/interrupt stack select (68020+)
- bits 10-8 I2/I1/I0: interrupt priority mask (0-7)
- A separate **supervisor stack pointer** A7' (or SSP). The user stack pointer is preserved as USP.
- The **vector base register** VBR. The exception vector table starts at the address in VBR. On reset VBR is 0.
- The source and destination function code registers SFC and DFC, used by the MOVES instruction.

- The cache control register CACR and cache address register CAAR.

User code that tries to access these registers raises a privilege violation exception (vector 8).

28.3 Memory model

Multi-byte values are stored **big-endian** in M68K main memory: the most-significant byte goes at the lowest address. A 32-bit write of 0x12345678 to address 0x1000 places 0x12 at 0x1000, 0x34 at 0x1001, 0x56 at 0x1002, 0x78 at 0x1003.

Intuition Engine's MMIO devices are little-endian, however; the bus normalises every MMIO access so an M68K `MOVE.L D0, ($F0800).L` delivers the same 32-bit value to the audio chip as an IE64 `STORE.L (R5), Rd`. Plain RAM accesses keep their natural big-endian order.

Word and long operands at odd addresses are valid on the 68020 (the 68000 would have raised an address error); IE follows the 68020 rule. Instruction fetches still require word alignment.

The address space is flat 32-bit. The MMIO map of Chapter 23 is accessible directly. The sign-extended alias at 0xFFFF0000 makes M68K's pre-decrement from zero (`-(SP)` with `SP = 0`) work.

28.4 Addressing modes

The 68020 provides 12 base modes plus several extensions:

#	Mode	Assembler syntax
1	Data register direct	Dn
2	Address register direct	An
3	Address register indirect	(An)
4	Postincrement	(An)+
5	Predecrement	-(An)
6	Indirect with 16-bit displacement	(d16,An)
7	Indirect with index + 8-bit displacement	(d8,An,Xn)
8	Absolute short	(xxx).W
9	Absolute long	(xxx).L
10	PC with 16-bit displacement	(d16,PC)
11	PC with index + 8-bit displacement	(d8,PC,Xn)
12	Immediate	#<data>
13	68020: memory indirect pre-indexed	([bd,An,Xn],od)
14	68020: memory indirect post-indexed	([bd,An],Xn,od)
15	68020: PC-relative memory indirect	([bd,PC,Xn],od)

Scaled indexing (`*1`, `*2`, `*4`, `*8`) is available on every mode that takes an Xn. The full-extension word adds an optional 32-bit base displacement and 32-bit outer displacement.

28.5 Instruction set

The full instruction set is in Appendix G. The groups below give a flavour of what is available; each mnemonic accepts a size suffix .B/.W/.L unless noted.

28.5.1 Data movement

MOVE, MOVEA (move into address register), MOVEQ (move quick - sign-extend an 8-bit immediate into a long), MOVEM (move multiple registers), EXG (exchange two registers), LEA (load effective address), PEA (push effective address), SWAP (swap halves of a long), LINK / UNLK (set up / tear down a stack frame), MOVES (68020+: move with function code).

28.5.2 Arithmetic

ADD, ADDA, ADDI, ADDQ, ADDX (add with extend); SUB, SUBA, SUBI, SUBQ, SUBX; MULU, Muls (32-bit on 68020+); DIVU, DIVS; NEG, NEGX; CLR; EXT and EXTB.L (sign-extend); CMP, CMPA, CMPI, CMPM, CMP2 (68020+: bounds-check compare). Decimal: ABCD, SBCD, NBCD, PACK, UNPK.

28.5.3 Logic

AND, ANDI, OR, ORI, EOR, EORI, NOT. The I-suffix forms accept an immediate as the source.

28.5.4 Shifts and rotates

ASL, ASR (arithmetic shifts); LSL, LSR (logical shifts); ROL, ROR (rotate); R0XL, R0XR (rotate through extend). Each can shift by a fixed 1-8 or by the count in a data register.

28.5.5 Bit operations and bit fields

BTST, BSET, BCLR, BCHG: test/set/clear/change one bit.

The 68020 adds full bit-field operations on arbitrary-width fields at arbitrary offsets in memory or registers: BFTST, BFEXTU, BFEXTS, BFCHG, BFCLR, BFSET, BFINS, BFFF0 (find first one).

28.5.6 Branches and jumps

Mnemonic	Effect
Bcc label	Conditional branch (Bcc = BEQ, BNE, BCS, BCC, BMI, BPL, BVS, BVC, BHI, BLS, BGT, BGE, BLT, BLE, BRA, BSR)
DBcc Dn, label	Decrement and branch if condition (Dn -= 1; branch if cc and Dn != -1)
Scc <ea>	Set byte to all-ones if condition true, else zero
JMP <ea>	Unconditional jump
JSR <ea>	Jump to subroutine
BSR label	Branch to subroutine (PC-relative)
RTS	Return from subroutine
RTD #n	Return and add n to SP

28.5.7 System

Mnemonic	Effect
NOP	No operation
STOP #imm	Set SR, stop until interrupt or trace
RESET	Assert the external reset line (peripherals reset)
TRAP #n	Software interrupt (vector 32+n)
TRAPV	Trap if overflow flag set
CHK <ea>, Dn	Trap if Dn outside bounds
CHK2	68020: bounds-check against a pair of bounds
MOVE to/from SR	Read/write status register (privileged on 68010+)
ANDI to SR, ORI to SR, EORI to SR	Bit operations on SR (privileged)
RTE	Return from exception
MOVEC	Read/write control register (privileged)
CAS, CAS2	Atomic compare-and-swap (68020+)
CALLM, RTM	Module call/return (68020+)

28.6 Exception model

The exception vector table holds 256 four-byte entries at the address in VBR. After reset $VBR = 0$, so the table starts at address 0.

Selected vector assignments (Appendix B of the M68K family programmer's reference):

Vector	Offset	Meaning
0	0x000	Reset: initial SSP
1	0x004	Reset: initial PC
2	0x008	Bus error
3	0x00C	Address error
4	0x010	Illegal instruction
5	0x014	Divide by zero
6	0x018	CHK / CHK2 instruction
7	0x01C	TRAPV instruction
8	0x020	Privilege violation
9	0x024	Trace
10	0x028	Line A trap (unimplemented A-line opcode)
11	0x02C	Line F trap (unimplemented F-line opcode)
24	0x060	Spurious interrupt

Vector	Offset	Meaning
25-31	0x064-0x07C	Auto-vectorized interrupt levels 1-7
32-47	0x080-0x0BC	TRAP #0 through TRAP #15
64+	0x100+	User-defined interrupts

When an exception fires, the CPU:

1. Computes a stack frame containing at least PC and SR.
2. Switches to supervisor mode if not already there.
3. Pushes the frame onto the supervisor stack.
4. Loads PC from $VBR + 4 * \text{vector}$.

RTE reverses the sequence and restores SR.

28.7 Interrupts on Intuition Engine

The Intuition Engine raises M68K interrupts at level 5 for video events (Copper, VBlank) and level 4 for audio engines, using auto-vector mode. A program that wants to respond to either installs a handler at the corresponding auto-vector slot:

```
move.l  #my_vbl_handler, $74      ; level-5 auto-vector
move.l  #my_aud_handler, $70      ; level-4 auto-vector
move    #$2000, sr                ; supervisor mode, mask < 5
```

Setting I2:I1:I0 in SR to 4 permits both; to 5 permits only level 5. Level 7 is non-maskable.

The IRQ diagnostics block (Chapter 23 §23.7) reports live counts of these auto-vectorized interrupts and is useful when debugging a hung handler.

28.8 What is not implemented

IE's 68020 implements 90%+ of the instruction set. The following are not in the chip:

- The MMU (PMMU, on-chip on the 68030).
- The coprocessor interface (cpGEN, cpScc, etc.) - FPU operations land directly through the x87-style IE FPU rather than through F-line coprocessor opcodes.
- Instruction cache modelling.
- Trace mode is decoded into T1/T0 but not enforced; the trace exception does not fire.
- Dynamic bus sizing - IE always assumes a 32-bit bus.

28.9 A small example

A 68K program that emits "HI" to the terminal and halts:

```
org      $1000

start:   move.l  #$F0700, a0          ; A0 = TERM_OUT
         move.b  #'H', (a0)
         move.b  #'I', (a0)
         move.b  #13, (a0)

         move.l  #$F07F0, a0         ; TERM_SENTINEL
         move.l  #$DEAD, (a0)        ; halt the CPU

halt:    bra     halt
```

Note `MOVE.B` to `(A0)` - the bus normalises the byte to the right endianness for the MMIO target.

28.10 What comes next

Chapter 29 covers the last of the heritage CPUs: x86, the family that originated as the 8086 in 1978 and went on to dominate the PC industry. Intuition Engine implements a 32-bit x86 processor with full 8086/8088 semantics plus the 386-era 32-bit register and operand extensions, run in a permanent flat real-address mode.